DGMS Examination 2019 — Question 59 Detailed Solution

Junior Scientific Officer (Geology) | Global Tectonics, Paleogeography & Himalayan Geology

QUESTION 59

The ocean that gave rise to birth of the Himalayan mountains was:

(a) Indian Ocean	(b) Pacific Ocean
(c) Tethys Ocean CORRECT ANSWER	(d) Panthalassa Ocean

1. Executive Summary & Answer Verification

The correct option is **(c) Tethys Ocean** (specifically the *Neo-Tethys Ocean*).

In global tectonics and historical geology (Eduard Suess, 1893; Dewey & Bird, 1970; Gansser, 1964):

- During the Mesozoic Era, the supercontinent Pangea was bifurcated into northern **Laurasia** (Eurasia and North America) and southern **Gondwanaland** (India, Africa, Antarctica, Australia, South America) by a vast, triangular equatorial seaway known as the **Tethys Ocean**.
- Following the Early Cretaceous breakup of Gondwana (~120 Ma), the Indian Plate broke loose and drifted rapidly northward across the Tethys at extraordinary velocities of 15–20 cm/year.
- Subduction of the Neo-Tethys oceanic lithosphere beneath the southern margin of Asia (the Lhasa terrane) generated the Gangdese batholith/volcanic arc. At approximately **50–55 Ma (Early Eocene)**, the ocean basin was completely consumed, initiating the continent-continent collision between India and Eurasia that crumpled the Tethyan marine sediments and Indian crustal margin into the **Himalayan Mountain Belt**.

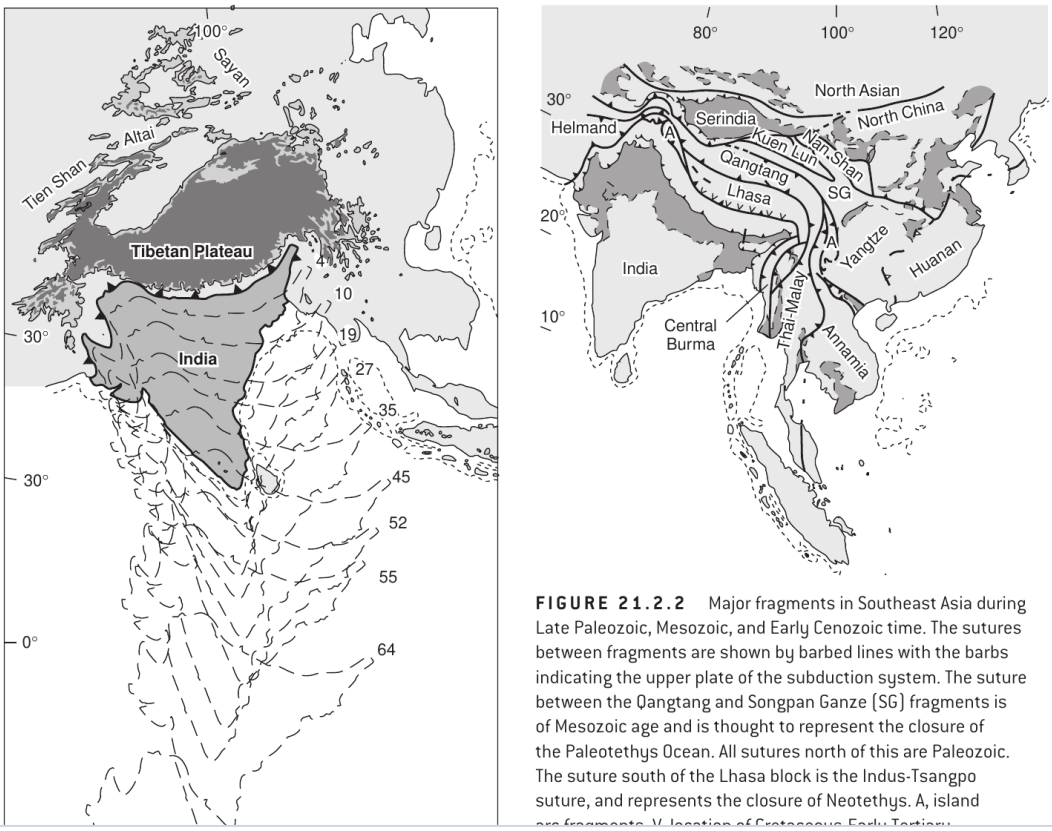


FIGURE 21.2.2 Major fragments in Southeast Asia during Late Paleozoic, Mesozoic, and Early Cenozoic time. The sutures between fragments are shown by barbed lines with the barbs indicating the upper plate of the subduction system. The suture between the Qangtang and Songpan Ganze (SG) fragments is of Mesozoic age and is thought to represent the closure of the Paleotethys Ocean. All sutures north of this are Paleozoic. The suture south of the Lhasa block is the Indus-Tsangpo suture, and represents the closure of Neotethys. A, island

Figure 59.1: *Northward drift of the Indian Plate closing the* ***Tethys Ocean*** *and colliding with the southern margin of Eurasia (Asia) from Late Cretaceous (71 Ma) through the Early Eocene (50 Ma) to the Present. Notice the sharp decrease in convergence rate from ~100 mm/yr to ~50 mm/yr at 50 Ma marking the initial continental collision. (Van der Pluijm & Marshak, Earth Structure, Figure 21.2.1, p. 526).*

2. Comprehensive Geological & Tectonic Evolution

2.1 Geological Record of the Tethys in the Himalayas

The closure of the Tethys Ocean is directly preserved across the tectonic zones of the Himalayas:

1. **Indus-Yarlung Tsangpo Suture Zone (IYTSZ):** The physical tectonic boundary where the Tethys Ocean disappeared. It contains dismembered ophiolites (fragments of Tethyan oceanic crust and upper mantle peridotites, such as the Spontang and Jungbwa ophiolites), deep-sea radiolarian cherts, and chaotic subduction melanges.
2. **Tethyan Sedimentary Sequence (Tethys Himalaya):** A continuous, highly fossiliferous marine sedimentary succession ranging in age from the Cambrian to the Eocene, deposited on the northern passive continental shelf of the Indian continent bordering the southern Tethys Ocean. Remarkably, the uppermost limestone cap of **Mount Everest (the Qomolangma Formation)** consists of Ordovician marine limestone deposited on the shallow floor of the Tethys Ocean, containing marine trilobites, brachiopods, and conodont fossils!
3. **Ammonite Fossils (Saligrams):** Black ammonite fossils (Jurassic *Ammonoidea*) found abundantly in the Spiti Shales of the Tethyan Himalaya (Muktinath, Nepal and Spiti, India) represent shallow-marine mollusks that swam in the Tethys Ocean before its destruction.

2.2 Chronology of Tethyan Evolution

Tectonic Phase	Geological Period	Plate Tectonic Event
Paleo-Tethys	Devonian – Late Triassic	Equatorial ocean between Gondwana and Laurasia; closed by Cimmerian collisions
Neo-Tethys Opening	Permian – Early Triassic	Rifting of Cimmerian microcontinents (Tibet, Iran, Turkey) away from northern Gondwana
Neo-Tethys Subduction	Jurassic – Late Cretaceous	Northward subduction beneath Eurasia, producing the Trans-Himalayan magmatic arc (Ladakh/Kohistan/Gangdese)
Terminal Ocean Closure	**Early Eocene (~50–55 Ma)**	**Final collision of India with Asia; birth of the Himalayan Mountain Chain**
Intracontinental Shortening	Oligocene – Present	Development of major Himalayan thrusts: MCT (Main Central Thrust), MBT (Main Boundary Thrust), and MFT (Main Frontal Thrust)

3. Evaluation of All Options

- **(a) Indian Ocean — INCORRECT:** The modern Indian Ocean was created *behind* India as it broke away from Antarctica and Australia; it did not close to form the Himalayas.
- **(b) Pacific Ocean — INCORRECT:** Located east of Asia and the Americas; unrelated to the collision of the Indian shield.
- **(c) Tethys Ocean — CORRECT:** The intervening ancient equatorial ocean whose subduction and collision produced the Himalayas and the Tibetan Plateau.
- **(d) Panthalassa Ocean — INCORRECT:** The vast global ancestral Pacific ocean that surrounded the entire supercontinent Pangea during the Paleozoic and Mesozoic.

Textbook Authority

Ben A. van der Pluijm & Stephen Marshak, *Earth Structure* (2nd ed., Chapter 21: Collisional Tectonics, pp. 525–526):
"A vast oceanic domain called Tethys formed a huge embayment into Pangea... The collision of India with Asia about 50 Ma closed the Tethys Ocean and produced the Himalaya."
P. K. Mukherjee, *Textbook of Geology*:
"The Himalayan mountains were formed out of a great geosynclinal sea known as the Tethys, which separated the northern landmass of Angaraland (Eurasia) from the southern Gondwanaland."